ECE 456 Problem Set 1:
The Basics of Electron Flow in Nanodevices

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1 Current with a Lorentzian Density of States

1.1 Alternative Form

Given the following equations for the number of channel electrons and current in a nanodevice

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1 f_1(E) + \gamma_2 f_2(E)}{\gamma_1 + \gamma_2} D(E - U)\,\mathrm{d}E \tag{1}$$

$$I = \frac{q}{\hbar} \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2} \int_{-\infty}^{\infty} [f_1(E) - f_2(E)] D(E - U)\,\mathrm{d}E \tag{2}$$

These equations can be manipulated to be easier to code into MATLAB. Firstly, change the variable of integration to $E' = E - U$. This can be manipulated to recieve

$$\begin{aligned} E' &= E - U \\ E &= E' + U \\ dE' &= dE \end{aligned}$$

Originally $E : -\infty \to \infty$, however, since subtracting a constant doesn't change infinity

$$E' : -\infty \to \infty$$

Hence, substituting for $E = E' + U$ and replacing $E' \to E$ produces

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1 f_1(E + U) + \gamma_2 f_2(E + U)}{\gamma_1 + \gamma_2} D(E)\,\mathrm{d}E \tag{3}$$

$$I = \frac{q}{\hbar} \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2} \int_{-\infty}^{\infty} [f_1(E + U) - f_2(E + U)] D(E)\,\mathrm{d}E \tag{4}$$

1.2 Solving the Equations Self-Consistently

Using the code in `problem1.py` to calculate the values of N and I self-consistently produced the plots seen in Figure 1.

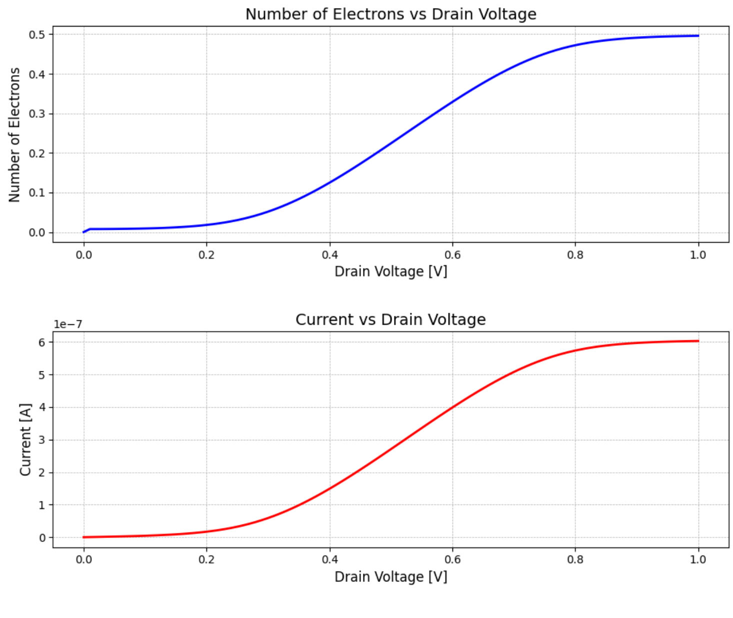

Figure 1: Plot of the current (I) and number of electrons (N) against the drain voltage (V_D)